| Reg. No | | | | | | | | | |
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**St. Joseph's College of Arts & Science (Autonomous)**
**St. Joseph's College Road, Cuddalore – 607001**
**B.Sc Degree Examination - APRIL 2021**

**CH306 - ANALYTICAL CHEMISTRY - II**

**Time : 3 hrs** **Max Marks :75**

**SECTION – A (20X1=20)**

Answer **ALL** Questions

**I. Choose the correct answer**

1. The technique used to study the change in weight of a substance with temperature over a period of time is
   a) TGA b) Gravimetric analysis c) Thermometric Titrations d) All the three
2. The precipitant used for estimating sulphate is
   a) $H_2SO_4$ b) $NH_4Cl$ c) KCl d) $BaCl_2$
3. Distillation under reduced pressure is carried out because
   a) The substances may get decomposed at its boiling point b) To get more amount of liquid
   c) It is cheap d) None of the above
4. In Steam Distillation
   a) A liquid which is immisible with water is distilled
   b) The immisible liquid distiles at a temperature lower than the boiling point of that liquid and water
   c) Steam is used to distile the liquid
   d) All the three
5. The cathode used in Polorography is
   a) A dropping mercury electrode b) Platinum c) Calomel d) Hydrogen
6. Polarimetry is used to measure the specific rotation of
   a) Glucose b) Acetic Acid c) Urea d) None of the above
7. Which of the following is a chromophore
   a) -N=N- b) $-NH_2$ c) -OH d) Cl
8. Which of the following is Bragg's equation
   a) $n\lambda=2d\sin\theta$ b) $n\lambda=2d\cos\theta$ c) $n\lambda=2d\sin\theta$ d) None of the above
9. Temporary hardness of water is due to
   a) bicarbonate of Ca b) Bicarbonate of Mg
   c) bicarbonates of Ca and Mg d) Chlorides of Ca and Mg
10. _____ is used in the estimation of Hardness of water
    a) Silver Nitrate b) EDTA c) potassium Permanganate d) Sodium Hydroxide

**II. Fill up the blanks**

11. The conversion of precipitate into colloidal solution by means of addition of a small amount of a suitable electrolyte is called _____
12. In the fractional crystalisation the compound with _____ solubility will crystallise first.
13. In Amperometric Titrations _____ plotted against the volume of Titrant added
14. Ethylene is an example for _____ electronic transition.
15. Desalination of sea water can be carried out by_____process._____2/-

**III. Match the following**

| | |
|---|---|
| 16. Gravimetric estimation of nickel | - Ion exchange resin |
| 17. Steam distillation | - Half wave potential |
| 18. Polarography | - NH2 |
| 19. Auxochrome | - DMG |
| 20. Demineralisation | - Aniline |

**SECTION – B (10X2=20)**

Answer any **TEN** Questions

21. What do you mean by Co-Precipitation? How it can be reduced?
22. What is DTA? What are the factors affecting DTA?
23. What is electroporosis?
24. What are selective precipitants? Give examples.
25. Give the advantages of DME?
26. The observed rotation is +18.6 for a solution which has a concentration of 15 gm in 100 ml solution in 20 cm long tube. Calculate the specific rotation.
27. What are the uses of polorography?
28. Give any two factors that affect the intensity of absorption maxima.
29. State Lambert -Beer law.
30. Write a note on BOD.
31. How is the temporary hardness of water is determined?
32. How salt water is purified by means of electrodialysis?

**SECTION- C (5X7=35)**

Answer any **FIVE** Questions

33. a) Sketch and explain the DGA curve of CuSO4 .5H2O
    b) What is post Precipitation?
34. Write notes on
    a) Fractional Distillation
    b) Steam Distillation
35. a) Explain how Acids of different strength compared making use of Polarimetry
    b) What is a polorogram?
36. a) Explain the terms involved in the Ilkovic equation.
    b) Explain the following.
    i) Concentration Polarisation ii) Half wave potential
37. Draw the Block diagram of the following and explain.
    a) Photo Colorimeter b) Spectro Photometer
38. Write notes on
    a) Zeolite Process b) Reverse osmosis
39. Explain in detail ion exchage process of demineralisation of water

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